



Theoretical modeling and characterization of equivalent magnetic properties in laminated composite magnetic shielding

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ABSTRACT

Compared to conventional magnetic shielding devices, the laminated composite shielding device composed of permalloy and nanocrystalline (PN-LCSD) has better static and low-frequency magnetic field shielding performance, ensuring the normal operation of ultrahigh-sensitivity sensors. However, the multi-thin layer characteristic of PN-LCSD presents significant challenges in evaluating its magnetic properties through analytical methods or simulation, which hinders the shielding performance analysis of PN-LCSD. Therefore, this paper proposes a theoretical model for the equivalent magnetic properties of PN-LCSD. Using the equivalent circuit method and geometric parameter analysis, the model derives equations for the equivalent permeability and shielding factor (SF) of PN-LCSD, resolving mesh-generation and convergence issues in performance analysis caused by its multi-thin layer characteristic and offering the analytical solution for its magnetic properties. Multiple experiments show that the maximum deviations between the analytical and experimental remanence and SF are only 6.48% and 5.78%, respectively. The model proposed in this study provides an effective theoretical support for the performance analysis and structural design of PN-LCSD.

1. Introduction

Devices including ultrasensitive comagnetometer, atomic spin inertial sensors, and biomedical magnetic signal detectors need to operate in near-zero magnetic field conditions [1–4]. Passive magnetic shielding is effective against static, alternating, and random magnetic disturbances, making it the most commonly used shielding method at present [5,6]. The basic principles of passive magnetic shielding include flux shunting through high-permeability materials and the eddy effect via high-conductivity materials [7]. Conventional shielding devices generally adopt a single-material strategy for the shielding layer, with permalloy for static and low-frequency magnetic field protection, and aluminum or copper for alternating magnetic field shielding [8,9].

Conventional shielding devices require the mutual nesting of different shielding layers to achieve lower residual magnetic induction, often resulting in complex structures [10]. Additionally, the performance of conventional shielding devices is also limited by the use of single-material shielding layers [11]. Permalloy exhibits relatively low saturation magnetization and its magnetic permeability rapidly decreases with frequency [12,13]. Meanwhile, aluminum and copper are unable to provide flux diversion, which poses challenges in enhancing

the shielding performance of the device in static and low-frequency magnetic fields [14]. Compared to permalloy, nanocrystalline offers significant advantages in terms of higher permeability, saturation flux density, and frequency stability [15,16]. However, nanocrystalline has a lower electrical conductivity, which results in a relatively weak eddy effect in alternating magnetic fields [17]. Additionally, nanocrystalline films require attachment to the supporting structure for use [18,19]. Xu et al. proposed a laminated composite shielding device composed of permalloy and nanocrystalline (PN-LCSD), which combines the magnetic advantages of nanocrystalline with the electrical advantages and mechanical strengths of permalloy [20].

Compared to conventional shielding devices, PN-LCSD has better static and low-frequency magnetic field shielding performance. Taking a PN-LCSD with a 2 mm shielding layer (1 mm 1J85 permalloy + 1 mm 1K107B nanocrystalline) and an internal space of 400 mm × 400 mm × 400 mm as an example, compared to the permalloy shielding device (2 mm 1J85 permalloy), the simulation shielding factor (SF) of PN-LCSD under static and 100 Hz magnetic fields increases by 24.47 % and 401.07 %, respectively. Moreover, compared to the permalloy-aluminum shielding device (1 mm 1J85 permalloy + 1 mm aluminum), the simulation SF of PN-LCSD under static and 100 Hz

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magnetic fields increases by 146.03 % and 281.04 %, respectively. PN-LCSD also offers advantages in cost and process. Unlike conventional shielding devices, which require complex multi-layer welding and nested assembly of permalloy or aluminum plates with fixed thickness series by skilled technicians, often leading to high costs, complex structure, and excessive weight. The nanocrystalline film attachment process of PN-LCSD can be completed by non-professionals, and the construction eliminates numerous welding and nesting processes, which makes PN-LCSD have lower cost and simpler process and structure than conventional shielding devices [21].

However, due to the multi-thin layer characteristic and magnetic circuit complexity of PN-LCSD, there is currently a lack of effective methods for calculating its equivalent magnetic properties, which poses challenges to its performance analysis [20]. Dou et al. proposed a method for calculating the equivalent permeability of the composite core used in wireless charging systems which is not suitable for PN-LCSD [22]. The composite core exhibits a discontinuous open magnetic structure, without considering the coupling effect between adjacent materials in its magnetic circuit. Moreover, its equivalent permeability calculation method is limited to cases where the magnetic field direction aligns with the core thickness and cannot be extended to calculate the shielding performance. In contrast, PN-LCSD features a continuous closed magnetic structure with varying magnetic circuits of its shielding layers at different positions, requiring case-specific analysis and consideration of magnetic circuit coupling effect between adjacent materials. Therefore, applying the calculation method of the composite magnetic core to analyze the magnetic properties of PN-LCSD is obviously unreasonable.

For the unique magnetic circuit structure of PN-LCSD, this paper proposes a novel theoretical model for the equivalent magnetic properties of PN-LCSD which can equate the composite shielding structure to a single-layer structure for analysis. The construction process of the model is shown in Fig. 1. The model can obtain the equivalent permeability of PN-LCSD, effectively resolving challenges in its performance analysis, including mesh generation and simulation convergence issues. Furthermore, the model provides the analytical solution for the SF of PN-

LCSD based on its equivalent permeability. Section 2 delineates the material magnetic testing and structural analysis of PN-LCSD. Section 3 provides the calculation method for the equivalent permeability of PN-LCSD along with the results of simulation validation. Section 4 deduces the calculation equations for SF applicable to PN-LCSD. Section 5 further substantiates the effectiveness of the proposed theoretical model through the experiment. Section 6 is the summary of the article. Section 7 further discusses the application prospect of the proposed model.

2. Magnetic measurement and structural analysis of PN-LCSD

2.1. Magnetic test and analysis of shielding materials

The magnetic properties of permalloy and nanocrystalline are crucial for the establishment of theoretical models for PN-LCSD. The soft magnetic instrument developed by Hunan Lian Zhong Technology, as shown in Fig. 2, is utilized to employ the simulated impact method and voltammetry for assessing the direct current (DC) and alternating current (AC) magnetic properties of 1J85 permalloy and 1K107B nanocrystalline, respectively. Details regarding the specific parameters of sample rings are shown in Table 1.

The DC magnetization curves of 1J85 and 1K107B are shown in Fig. 3. The initial magnetic permeability and saturation magnetization of 1K107B are higher than those of 1J85. The amplitude permeability curves of 1J85 and 1K107B are shown in Fig. 4. The amplitude permeability of 1J85 sharply decreases with increasing frequency. Specifically, at a frequency of 100 Hz, the initial amplitude permeability of 1J85 drops to 9.35k, which is only 18.87 % of the initial permeability under the static magnetic field. In contrast, the amplitude permeability of 1K107B at 100 Hz is more than 70k, which has high-frequency stability. Therefore, the nanocrystalline should be attached to the outside of permalloy to achieve better shielding effect.

2.2. Structural analysis and characterization of PN-LCSD

The composite shielding layer structure of PN-LCSD is shown in Fig. 5(a), where the nanocrystalline film is systematically layered onto the surface of the permalloy sheet. This configuration promotes a path of low impedance for the magnetic flux, while the permalloy substrate ensures the structural stability of the composite shielding layer. The films are connected by an insulating paste to avoid the introduction of additional magnetic field interference and to alleviate the brittle defects

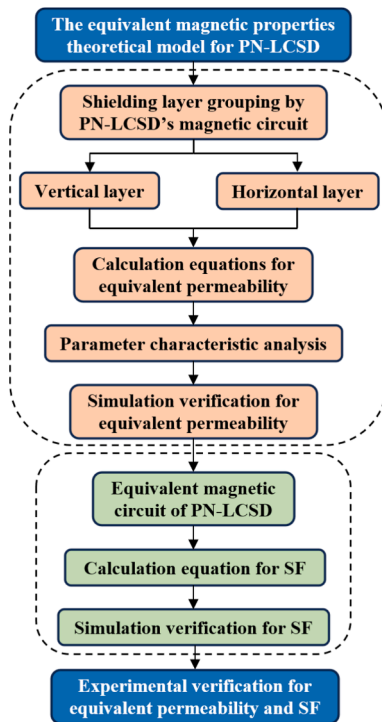


Fig. 1. The construction flowchart of the equivalent magnetic properties theoretical model for PN-LCSD.

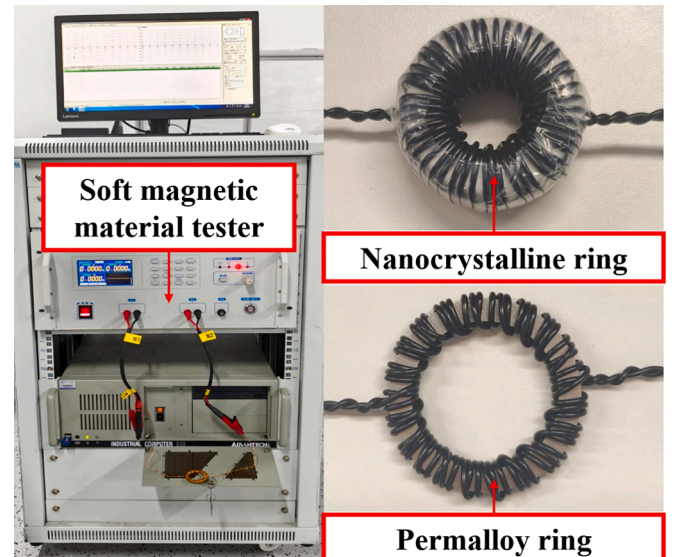


Fig. 2. Soft magnetic material tester and sample rings.

Table 1

The sample ring parameters of 1J8 5 permalloy and 1K107B nanocrystalline for magnetic test.

Parameters	1J85	1K107B
Outer diameter D (mm)	40	30
Inner diameter d (mm)	30	20
Thickness t (mm)	2	10
Quality m (g)	9.18	30
Stacking factor	1	0.82
The primary winding turns N_1	20	17
The second winding turns N_2	40	31

of the nanocrystalline film after annealing. The complex thin-layer structure of PN-LCSD makes theoretical analysis and simulation difficult. For engineering applications, the non-magnetic properties and thin-layer characteristics of the insulating paste allow its influence on the composite shielding layer to be approximately neglected. Additionally, when the direction of the external magnetic field is parallel to the thickness direction of the n -layer nanocrystalline structure, as shown in Fig. 6(a), nanocrystalline films are connected in series with each other in the magnetic circuit. The total magnetoresistance (the magnetic resistance of material) R_S of the n -layer nanocrystalline structure in series magnetic circuit can be expressed as:

$$\begin{cases} R_S = R_{S1} + R_{S2} + \dots + R_{Sn} = \frac{h_N}{l_N w_N \mu_N} \\ R_{S1} = R_{S2} = \dots = R_{Sn} = \frac{h_N}{n l_N w_N \mu_N} \end{cases} \quad (1)$$

where R_{S1} to R_{Sn} are the magnetoresistance of the 1st to n layer nanocrystalline film in the series circuit, l_N and w_N are the length and width of the nanocrystalline film, h_N is the thickness of the n -layer

nanocrystalline structure, and μ_N is the permeability of the nanocrystalline.

When the direction of the external magnetic field is perpendicular to the thickness direction of the n -layer nanocrystalline structure, as shown in Fig. 6(b), nanocrystalline films are connected in parallel with each other in the magnetic circuit. The total magnetoresistance R_P of the n -layer nanocrystalline structure in parallel magnetic circuit can be expressed as:

$$\begin{cases} R_P = \left(\frac{1}{R_{P1}} + \frac{1}{R_{P2}} + \dots + \frac{1}{R_{Pn}} \right)^{-1} = \frac{l_N}{w_N h_N \mu_N} \\ R_{P1} = R_{P2} = \dots = R_{Pn} = \frac{n l_N}{w_N h_N \mu_N} \end{cases} \quad (2)$$

where R_{P1} to R_{Pn} are the magnetoresistance of the 1st to n layer nanocrystalline film in the parallel circuit. Based on Eq. (1) and Eq. (2), the multilayer nanocrystalline structure can be analyzed as a whole magnet in the magnetic circuit. Therefore, as shown in Fig. 5(b), the composite shielding layer of PN-LCSD can be conceptually simplified to a bilayer structure for analysis.

3. Permeability analytical model of PN-LCSD

The PN-LCSD comprises six composite shielding layers. The composite shielding layer where the direction of the magnetic field is aligned with the thickness direction is defined as vertical layer (VL), and the composite shielding layer where the direction of the magnetic field is perpendicular to the thickness direction is defined as horizontal layer (HL), as shown in Fig. 7.

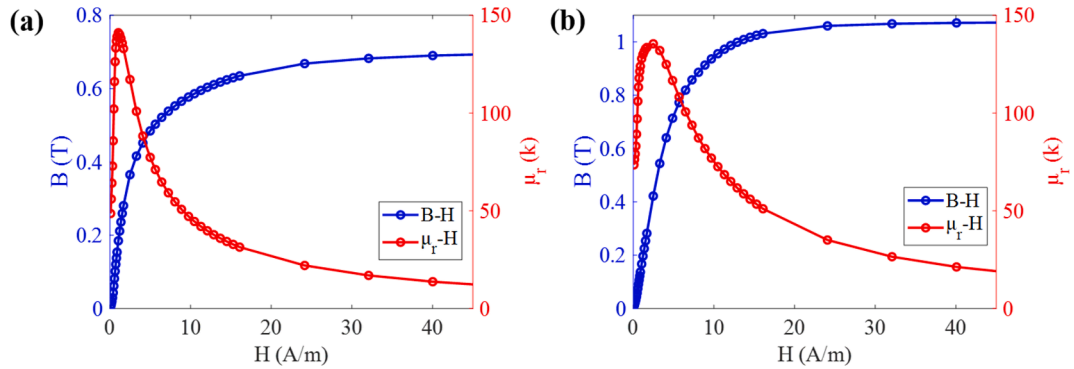


Fig. 3. DC magnetization curves. (a) 1J85, (b) 1K107B.

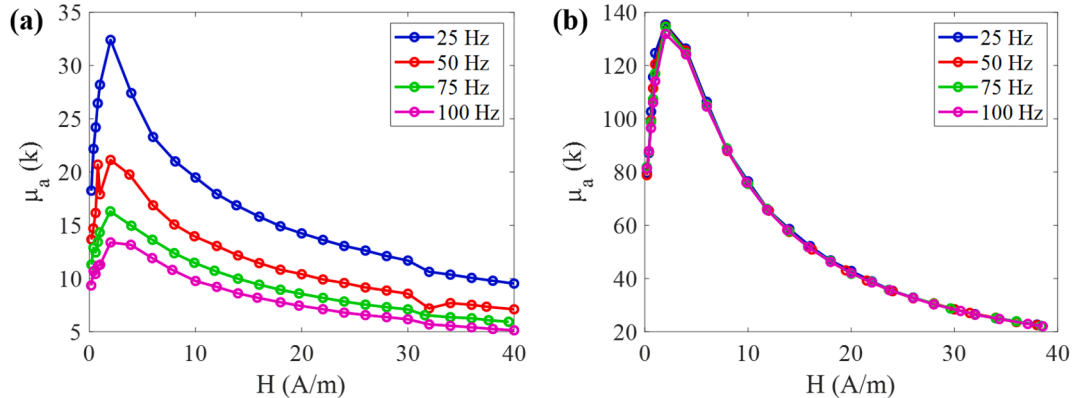


Fig. 4. Amplitude permeability curves. (a) 1J85, (b) 1K107B.

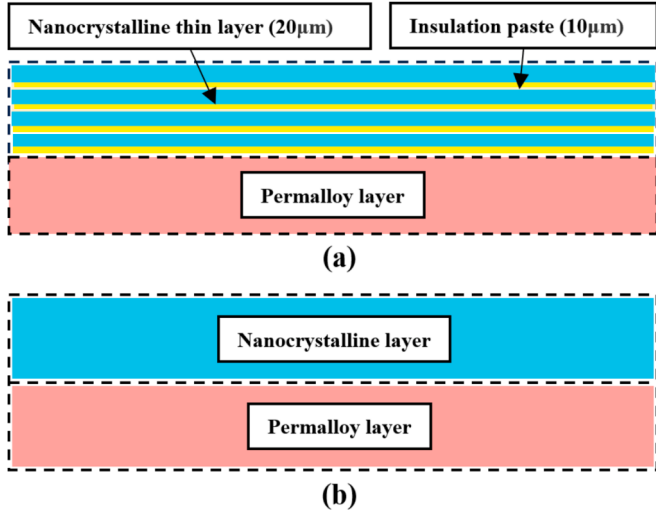


Fig. 5. The composite shielding layer structure of PN-LCSD. (a) Actual structure, (b) simplified structure.

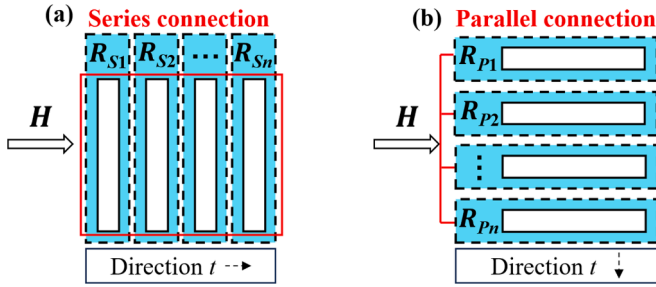


Fig. 6. Magnetic circuit of n -layer nanocrystalline structure. (a) Series connection, (b) parallel connection.

3.1. Construction of equivalent permeability calculation model

The magnetic circuit corresponding to VL is shown in Fig. 8(a) and the nanocrystalline and permalloy are placed at the bottom and top layers, respectively. Within the three-dimensional coordinate system, R_{Bx} and R_{Tx} represent the magnetoresistance of the bottom and top materials along the x direction, respectively. R_{By} and R_{Ty} represent the magnetoresistance of the bottom and top materials along the y direction, respectively. R_{Bz} and R_{Tz} represent the magnetoresistance of the bottom and top materials along the z direction, respectively. R_B and R_T represent the total magnetoresistance of the bottom and top materials, respectively. The magnetoresistance of the bottom and top materials can be expressed as:

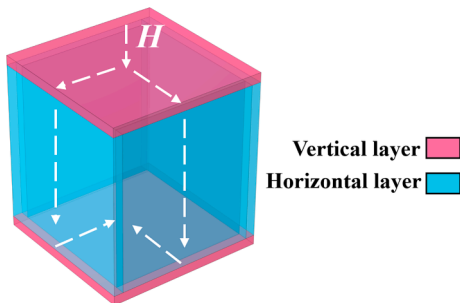


Fig. 7. The division of the vertical layer and horizontal layer.

$$\begin{cases} R_{Bx} = \frac{l_B}{\mu_{Bx} w_B d_B} R_{Tx} = \frac{l_T}{\mu_{Tx} w_T d_T} \\ R_{By} = \frac{w_B}{\mu_{By} l_B d_B} R_{Ty} = \frac{w_T}{\mu_{Ty} l_T d_T} \\ R_{Bz} = \frac{d_B}{\mu_{Bz} l_B w_B} R_{Tz} = \frac{d_T}{\mu_{Tz} l_T w_T} \\ R_B = \frac{R_{Bx} R_{By} R_{Bz}}{R_{Bx} R_{By} + R_{Bx} R_{Bz} + R_{By} R_{Bz}} \\ R_T = \frac{R_{Tx} R_{Ty} R_{Tz}}{R_{Tx} R_{Ty} + R_{Tx} R_{Tz} + R_{Ty} R_{Tz}} \end{cases} \quad (3)$$

where l_B , w_B and d_B are the length, width and thickness of the bottom layer, respectively. l_T , w_T and d_T are the length, width and thickness of the top layer, respectively. μ_{Bx} , μ_{By} and μ_{Bz} are the permeabilities of the bottom layer in the x , y , and z directions, respectively. μ_{Tx} , μ_{Ty} and μ_{Tz} are the permeabilities of the top layer in the x , y , and z directions, respectively. According to Fig. 8(b) and the series-parallel theory of magnetoresistance, the total magnetoresistance R_V of VL can be calculated as:

$$R_V = \frac{R_{Bx} R_{By} (R_{Bz} + R_T)}{R_{Bx} R_{By} + R_{Bx} (R_{Bz} + R_T) + R_{By} (R_{Bz} + R_T)} \quad (4)$$

In accordance with the principle of magnetic flux conservation, the magnetic flux components of VL in the x , y and z directions satisfy:

$$\begin{cases} \varphi_{Vx} = \varphi_{Bx} + \varphi_{Tx} \\ \varphi_{Vy} = \varphi_{By} + \varphi_{Ty} \\ \varphi_{Vz} = \varphi_{Bz} + \varphi_{Tz} \end{cases} \quad (5)$$

where φ_{Bx} , φ_{Tx} , φ_{By} , φ_{Ty} , φ_{Bz} and φ_{Tz} are the outgoing magnetic flux components of the bottom and top materials in the x , y and z directions. From the magnetomotive force theory and Kirchhoff's law of magnetic circuits, the outgoing magnetic flux components of the bottom and top materials can be written as follows:

$$\begin{cases} \varphi_{Bx} = \frac{R_V \varphi_V}{R_{Bx}} \varphi_{By} = \frac{R_V \varphi_V}{R_{By}} \varphi_{Bz} = \frac{R_V \varphi_V}{R_{Bz} + R_T} \\ \varphi_{Tx} = \frac{R_T \varphi_{Bz}}{R_{Tx}} \varphi_{Ty} = \frac{R_T \varphi_{Bz}}{R_{Ty}} \varphi_{Tz} = \frac{R_T \varphi_{Bz}}{R_{Tz}} \end{cases} \quad (6)$$

where φ_V is the total inflow flux of VL. Considering the VL as a whole, the magnetoresistance of VL along the x , y and z directions can be calculated as:

$$\begin{cases} R_{Vx} = \frac{R_V \varphi_V}{\varphi_{Vx}} \\ R_{Vy} = \frac{R_V \varphi_V}{\varphi_{Vy}} \\ R_{Vz} = \frac{R_V \varphi_V}{\varphi_{Vz}} \end{cases} \quad (7)$$

Substituting Eq. (3) to Eq. (6) into Eq. (7), the magnetoresistance of VL along the x , y and z directions can be calculated as:

$$\begin{cases} R_{Vx} = \frac{R_{Bx} R_{Tx} (R_{Bz} + R_T)}{R_{Bx} R_{Tx} + R_{Tx} (R_{Bz} + R_T)} \\ R_{Vy} = \frac{R_{By} R_{Ty} (R_{Bz} + R_T)}{R_{By} R_{Ty} + R_{Ty} (R_{Bz} + R_T)} \\ R_{Vz} = \frac{R_{Tz} (R_{Bz} + R_T)}{R_T} \end{cases} \quad (8)$$

According to Fig. 8(a), $l = (l_B + l_T) / 2$, $w = (w_B + w_T) / 2$ and $d = d_B + d_T$ represent the length, width and thickness of the composite shielding layer, respectively. Therefore, the equivalent permeability equation of VL can be written as:

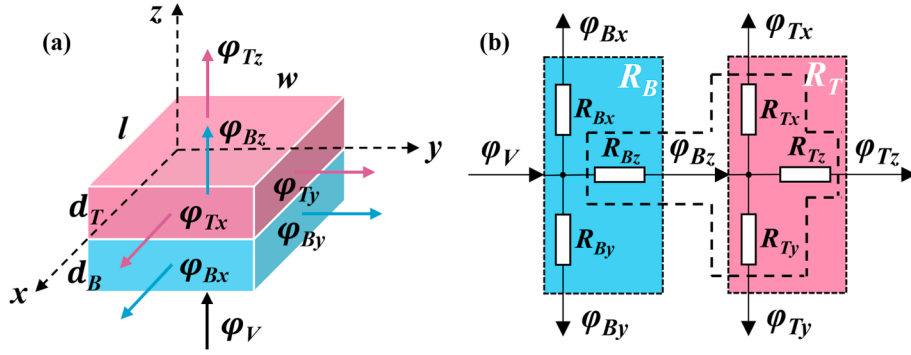


Fig. 8. Magnetic circuit of VL. (a) Circuit diagram, (b) magnetoresistance relationship.

$$\begin{cases} \mu_{Vx} = \frac{l}{wdR_{Vx}} = \frac{(l_B + l_T)}{(w_B + w_T)(d_B + d_T)} \left[\frac{R_{Bx}R_T + R_{Tx}(R_{Bz} + R_T)}{R_{Bx}R_{Tx}(R_{Bz} + R_T)} \right] \\ \mu_{Vy} = \frac{w}{ldR_{Vy}} = \frac{(w_B + w_T)}{(l_B + l_T)(d_B + d_T)} \left[\frac{R_{By}R_T + R_{Ty}(R_{Bz} + R_T)}{R_{By}R_{Ty}(R_{Bz} + R_T)} \right] \\ \mu_{Vz} = \frac{d}{lwR_{Vz}} = \frac{4(d_B + d_T)}{(l_B + l_T)(w_B + w_T)} \left[\frac{R_T}{R_{Tz}(R_{Bz} + R_T)} \right] \end{cases} \quad (9)$$

where μ_{Vx} , μ_{Vy} and μ_{Vz} are the permeabilities of VL in the x , y , and z directions. The magnetic circuit corresponding to HL is shown in Fig. 9 (a) and the nanocrystalline and permalloy are placed at the bottom and top layers, respectively.

The incident magnetic flux components ϕ_B and ϕ_T of the bottom and top materials in HL satisfy:

$$\begin{cases} \phi_B = \frac{R_T \phi_H}{R_B + R_T} \\ \phi_T = \frac{R_B \phi_H}{R_B + R_T} \end{cases} \quad (10)$$

where ϕ_H is the total inflow flux of HL. According to Fig. 9(b), the total magnetoresistance R_H of HL can be calculated as:

$$R_H = \frac{R_B R_T}{R_B + R_T} \quad (11)$$

According to the magnetic flux theory, the outgoing magnetic flux of HL in the x , y and z directions can be expressed as:

$$\begin{cases} \phi_{Hx} = \phi_{Bx} + \phi_{Tx} = \frac{R_B \phi_B}{R_{Bx}} + \frac{R_T \phi_T}{R_{Tx}} \\ \phi_{Hy} = \phi_{By} + \phi_{Ty} = \frac{R_B \phi_B}{R_{By}} + \frac{R_T \phi_T}{R_{Ty}} \\ \phi_{Hz} = \frac{R_H \phi_H}{R_{Bz} + R_{Tz}} \end{cases} \quad (12)$$

Considering the HL as a whole, the magnetoresistance of HL in the x , y and z directions can be expressed as:

$$\begin{cases} R_{Hx} = \frac{R_H \phi_H}{\phi_{Hx}} \\ R_{Hy} = \frac{R_H \phi_H}{\phi_{Hy}} \\ R_{Hz} = \frac{R_H \phi_H}{\phi_{Hz}} \end{cases} \quad (13)$$

Substituting Eq. (3), Eq. (10), Eq. (11) and Eq. (12) into Eq. (13), the magnetoresistance of HL along the x , y and z directions can be obtained as:

$$\begin{cases} R_{Hx} = \frac{R_{Bx} R_{Tx}}{R_{Bx} + R_{Tx}} \\ R_{Hy} = \frac{R_{By} R_{Ty}}{R_{By} + R_{Ty}} \\ R_{Hz} = R_{Bz} + R_{Tz} \end{cases} \quad (14)$$

Therefore, the equivalent permeability equation of HL can be expressed as:

$$\begin{cases} \mu_{Hx} = \frac{l}{wdR_{Hx}} = \frac{(l_B + l_T)}{(w_B + w_T)(d_B + d_T)} \left(\frac{R_{Bx} + R_{Tx}}{R_{Bx} R_{Tx}} \right) \\ \mu_{Hy} = \frac{w}{ldR_{Hy}} = \frac{(w_B + w_T)}{(l_B + l_T)(d_B + d_T)} \left(\frac{R_{By} + R_{Ty}}{R_{By} R_{Ty}} \right) \\ \mu_{Hz} = \frac{d}{lwR_{Hz}} = \frac{4(d_B + d_T)}{(l_B + l_T)(w_B + w_T)} \left(\frac{1}{R_{Bz} + R_{Tz}} \right) \end{cases} \quad (15)$$

where μ_{Hx} , μ_{Hy} and μ_{Hz} are the permeabilities of HL in the x , y , and z directions. Therefore, the equivalent permeability in three directions for PN-LCSD can be calculated using Eq. (3), Eq. (9) and Eq. (15).

3.2. Characteristic analysis of equivalent permeability calculation model

When $d_B = d_T = 2$ mm, the variation in the equivalent permeability of VL and HL in three directions with l and w is shown in Fig. 10. The

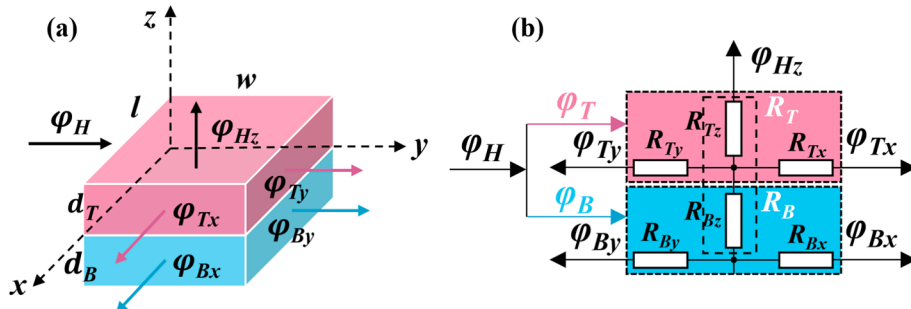


Fig. 9. Magnetic circuit of HL. (a) Circuit diagram, (b) magnetoresistance relationship.

variation in the equivalent permeability of VL and HL in three directions is minimal with l and w in the range of 100 to 2000 mm and the influence of l and w on the equivalent permeability can be ignored. The variation in the equivalent permeability with d_B and d_T when $l = w = 400$ mm is shown in Fig. 11. The permeability variation of VL and HL with respect to d_B and d_T appears to be complex. However, when the thickness ratio $p = d_B / d_T$ is fixed, the equivalent permeability of the composite shielding layer becomes independent of the specific values of d_B and d_T . Therefore, p is the decisive factor of the equivalent permeability for PN-LCSD, which characterizes the proportion of nanocrystalline and permalloy in the composite shielding layer.

The variation in the equivalent permeability with p is shown in Fig. 12, when p gradually increases, the equivalent permeability of VL and HL transitions from 1J85 to 1K107B. Given that the composite shielding layer consists of uniformly shaped plates and that the effects of l and w on the equivalent permeability are negligible, it can be assumed that $l_B = l_T = l$ and $w_B = w_T = w$. Substituting Eq. (3), Eq. (9) and Eq. (15), the equivalent permeability equation can be obtained as:

$$\left\{ \begin{array}{l} \mu_{Vx} = \mu_{Vy} = \frac{l^2 w^2 d_T^2 \mu_B \mu_T + (l^2 w^2 + l^2 d_T^2 + w^2 d_T^2) d_B^2 \mu_B \mu_T + l^2 w^2 d_B d_T \mu_B^2}{[(l^2 w^2 + l^2 d_T^2 + w^2 d_T^2) d_B \mu_T + l^2 w^2 d_T \mu_B] (d_B + d_T)} \\ \mu_{Vz} = \frac{l^2 w^2 d_T \mu_B \mu_T^2 (d_B + d_T)}{(l^2 w^2 + l^2 d_T^2 + w^2 d_T^2) d_B d_T \mu_T^2 + l^2 w^2 d_T^2 \mu_B \mu_T} \\ \mu_{Hx} = \mu_{Hy} = \frac{d_B \mu_B + d_T \mu_T}{d_B + d_T} \\ \mu_{Hz} = \frac{(d_B + d_T) \mu_B \mu_T}{d_B \mu_T + d_T \mu_B} \end{array} \right. \quad (16)$$

where μ_B represents the permeability of nanocrystalline, and μ_T represents the permeability of permalloy. Since d_B and d_T of the composite shielding layer is much smaller than l and w , Eq. (16) can be simplified as:

$$\left\{ \begin{array}{l} \mu_{Vx} = \mu_{Vy} = \frac{(1 + p^2) \mu_B \mu_T + p \mu_B^2}{(1 + p) (\mu_B + p \mu_T)} \\ \mu_{Hx} = \mu_{Hy} = \frac{p \mu_B + \mu_T}{1 + p} \\ \mu_{Vz} = \mu_{Hz} = \frac{(1 + p) \mu_B \mu_T}{\mu_B + p \mu_T} \end{array} \right. \quad (17)$$

3.3. Verification of equivalent permeability calculation model

According to Eq. (17), p of 0.5 ($d_B = 1$ mm, $d_T = 2$ mm), 1 ($d_B = d_T = 2$ mm), 1.5 ($d_B = 1.5$ mm, $d_T = 1$ mm) and 2 ($d_B = 2$ mm, $d_T = 1$ mm) correspond to the equivalent permeability of PN-LCSD are shown in Table 2. Utilizing COMSOL software, the layered model (L-model) and the equivalent model (E-model) are established according to the test permeability and equivalent permeability respectively. The remanence (the residual magnetic induction) simulation analyses are conducted on L-model and E-model, both featuring internal dimension of 400 mm $\times$ 400 mm $\times$ 400 mm. The internal remanence distribution of L-model and E-model is basically the same, as shown in Fig. 13.

The comparison of average remanence B_r in the central region (200 mm $\times$ 200 mm $\times$ 200 mm) between L-model and E-model is presented in Fig. 14(a), and the maximum deviation of B_r is merely 1.60 %. The variation trends presented in Fig. 14(a) are further analyzed as follows. Under the static magnetic field, the B_r of PN-LCSD decreases with the increase in both the shielding layer thickness and p , which explains the changing trend of B_r shown in Fig. 14(a). Moreover, as the difficulty of mesh generation and convergence of the PN-LCSD simulation model decreases with the increase in the shielding layer thickness, a negative correlation trend emerges between the B_r deviation of L-model and E-model and the shielding layer thickness of PN-LCSD. This explains the fluctuation trend of the B_r deviation depicted in Fig. 14(a).

In addition to static magnetic fields, PN-LCSD also require operation within low-frequency magnetic fields. The L-model and E-model,

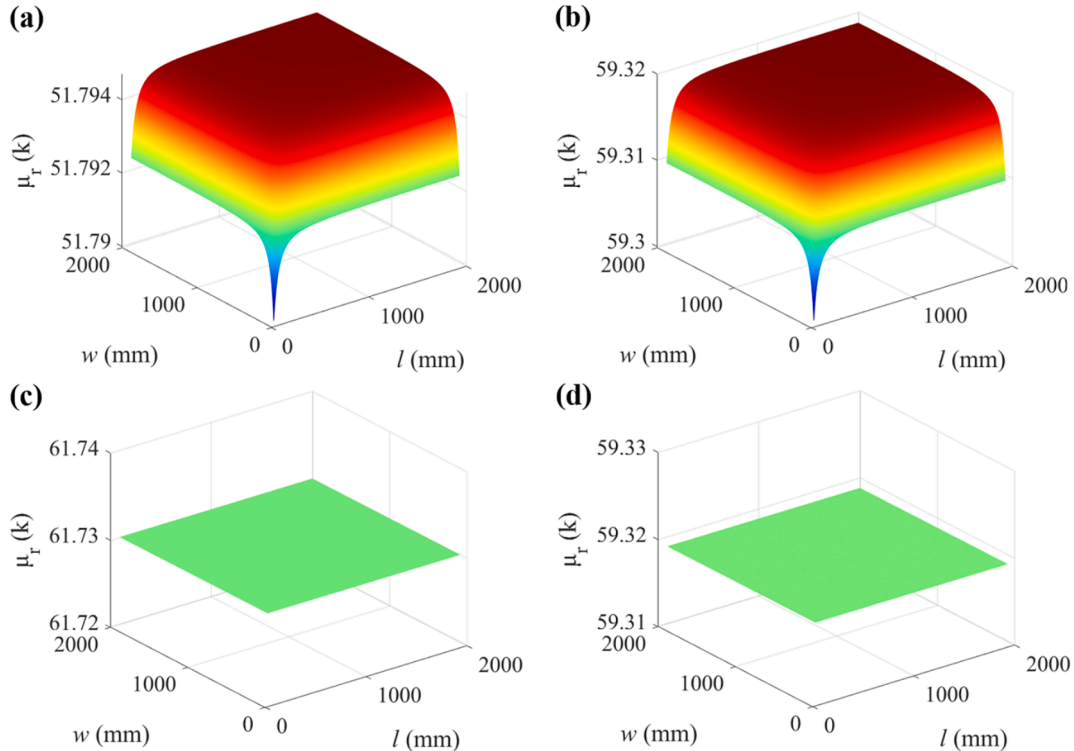


Fig. 10. The variation of the equivalent permeability with l and w . (a) The x and y directions of VL, (b) the z direction of VL, (c) the x and y directions of HL, (d) the z direction of HL.

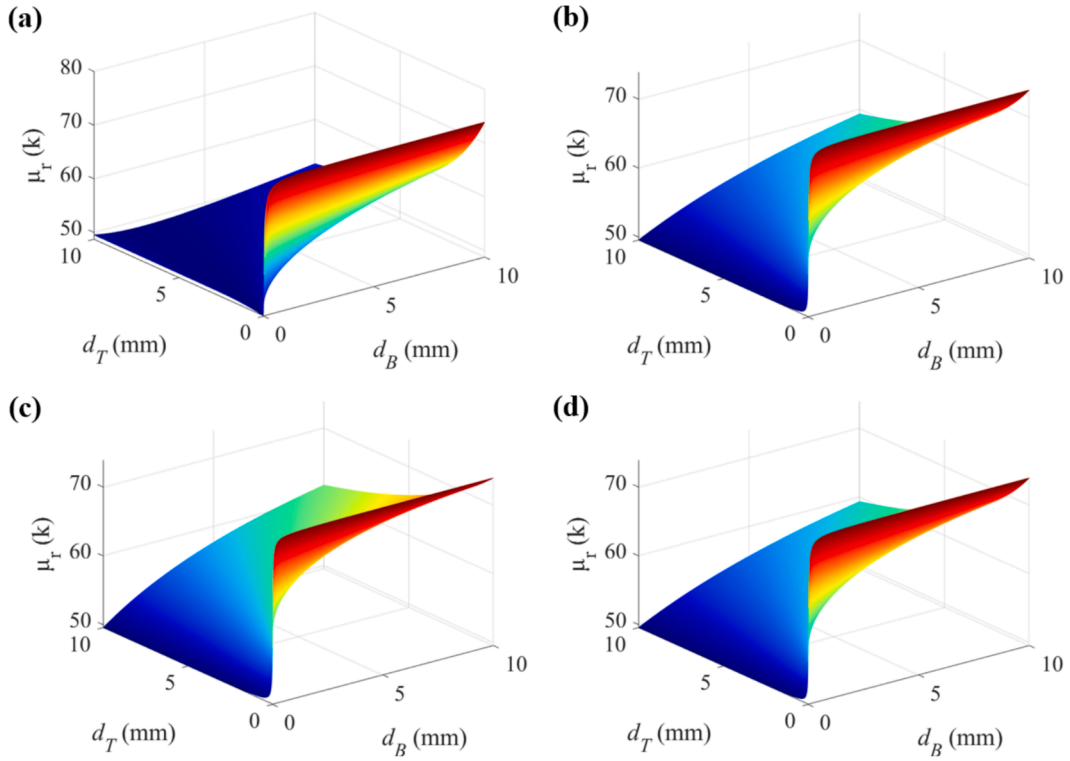


Fig. 11. The variation of the equivalent permeability with d_B and d_T . (a) The x and y directions of VL, (b) the z direction of VL, (c) the x and y directions of HL, (d) the z direction of HL.

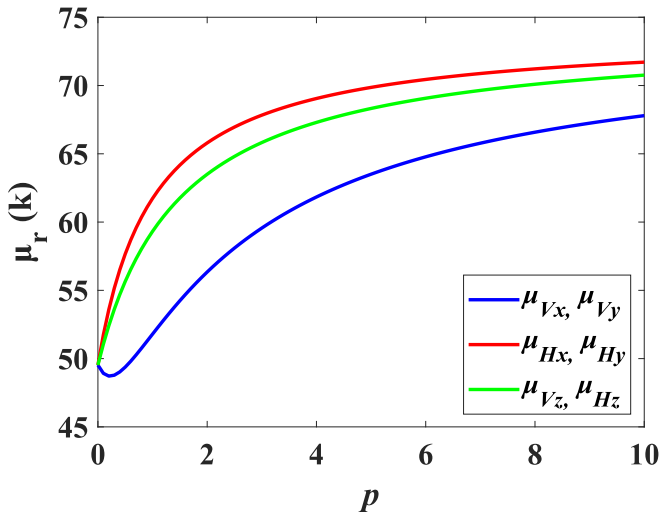


Fig. 12. The variation of the equivalent permeability for VL and HL with p .

Table 2

The equivalent permeability of simulation models for PN-LCSD at different thickness ratios.

p	μ_{Vx} (k)	μ_{Vy} (k)	μ_{Vz} (k)	μ_{Hx} (k)	μ_{Hy} (k)	μ_{Hz} (k)
0.5	49.38	49.38	55.65	57.66	57.66	55.65
1	51.79	51.79	59.32	61.73	61.73	59.32
1.5	54.24	54.24	61.76	64.17	64.17	61.76
2	56.34	56.34	63.50	65.80	65.80	63.50

corresponding to $p = 1$, are subjected to simulation analysis under alternating magnetic fields at frequencies of 25 Hz, 50 Hz, 75 Hz, and 100 Hz, respectively. To remove the interference caused by the eddying

effect, electrical conductivity is set to zero. According to Eq. (17) and the initial amplitude permeabilities of 1J85 and 1K107B, the equivalent permeabilities of E-model at frequencies of 25 Hz, 50 Hz, 75 Hz, and 100 Hz are shown in Table 3.

The internal remanence distribution obtained from the simulation is shown in Fig. 15, which demonstrates a high level of consistency between L-model and E-model. The comparison of B_r for L-model and E-model under the alternating magnetic field is presented in Fig. 14(b), and the maximum deviation of B_r is merely 0.57 %. The simulation results validate the accuracy and effectiveness of the equivalent permeability calculation model.

4. Shielding performance analytical model of PN-LCSD

The SF is the key indicator for evaluating the performance of the magnetic shielding device. Based on the equivalent permeability, the equivalent magnetic circuit of PN-LCSD is shown in Fig. 16. The total magnetic flux φ_0 , the magnetic flux within the composite shielding layer φ_S , and the magnetic flux in the shielding space φ_I can be expressed as follows:

$$\begin{cases} \varphi_0 = \mu_0 H_0 a \\ \varphi_S = 2\mu_{Hx} H_S t \\ \varphi_I = \mu_0 H_I (a - 2t) \end{cases} \quad (18)$$

where μ_0 is the vacuum permeability, H_0 is the external magnetic field intensity, H_I is the internal magnetic field intensity, a is the side length of the rectangular section, and t is the thickness of the composite shielding layer. Since $\varphi_0 = \varphi_S + \varphi_I$ and according to Eq. (18), it follows that:

$$\mu_0 H_0 a = 2\mu_{Hx} H_S t + \mu_0 H_I (a - 2t) \quad (19)$$

For the magnetic circuit C_S , the magnetomotive force U_S can be expressed as:

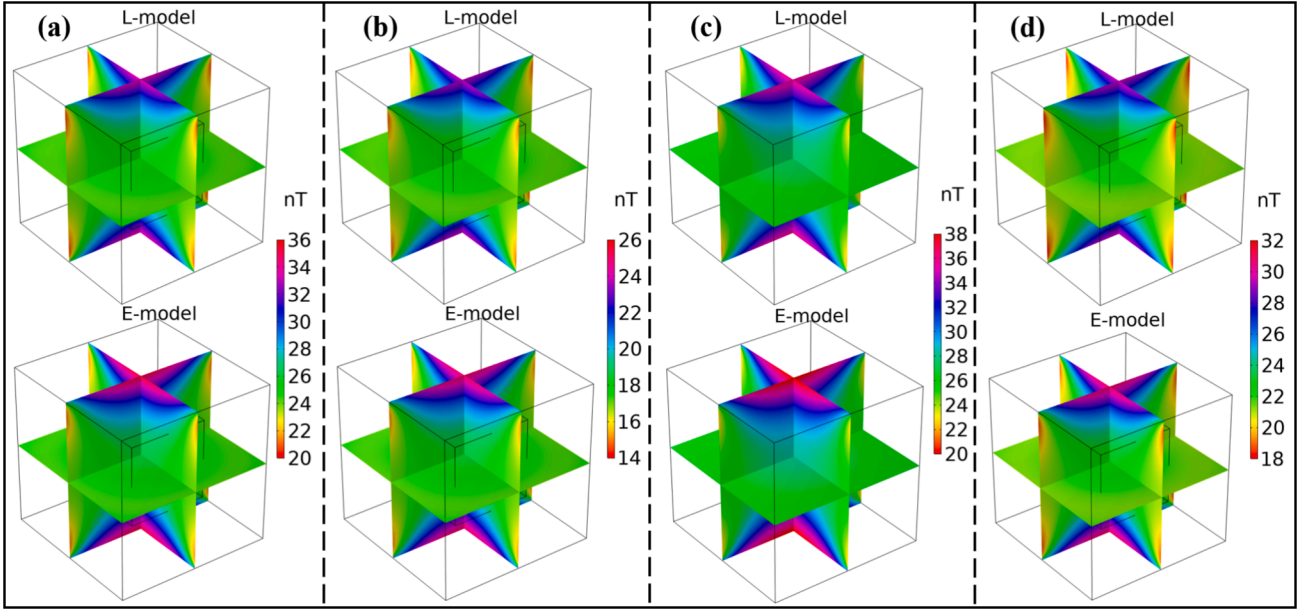


Fig. 13. The distribution of remanence for L-model compared to E-model with different p . (a) $p = 0.5$, (b) $p = 1$, (c) $p = 1.5$, (d) $p = 2$.

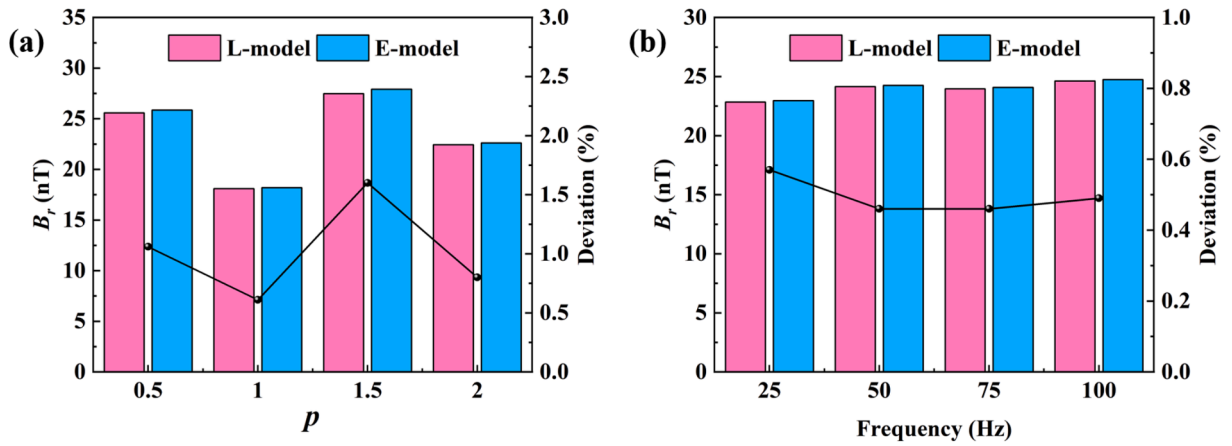


Fig. 14. Comparison of B_r between L-model and E-model. (a) Different p , (b) different frequency.

Table 3

The equivalent permeability of simulation models for PN-LCSD at different frequencies.

Frequency (Hz)	μ_{vx} (k)	μ_{vy} (k)	μ_{vz} (k)	μ_{Hx} (k)	μ_{Hy} (k)	μ_{Hz} (k)
25	47.22	47.22	29.50	48.90	48.90	29.50
50	45.26	45.26	23.14	46.26	46.26	23.14
75	45.93	45.93	19.73	46.60	46.60	19.72
100	44.88	44.88	16.64	45.35	45.35	16.64

$$\begin{cases} U_S = 2U_{Sx} + U_{Sz} \\ U_{Sx} = \frac{R_{Sx}\varphi_S}{2} \\ U_{Sz} = \frac{R_{Sz}\varphi_S}{2} \end{cases} \quad (20)$$

where R_{Sx} is the magnetoresistance of C_S along the x direction, R_{Sz} is the magnetoresistance of C_S along the z direction, and b is the height of PN-LCSD. The R_{Sx} and R_{Sz} can be calculated as:

$$\begin{cases} R_{Sx} = \frac{a}{2\mu_{vx}t} \\ R_{Sz} = \frac{b-2t}{\mu_{Hx}t} \end{cases} \quad (21)$$

Similarly, for the magnetic circuit C_I , the magnetomotive force U_I can be expressed as:

$$\begin{cases} U_I = 2U_{Iz} + U_{IS} \\ U_{Iz} = R_{Iz}\varphi_I \\ U_{IS} = R_{IS}\varphi_I \end{cases} \quad (22)$$

where R_{Iz} is the magnetoresistance of C_I along the z direction, and R_{IS} is the internal space magnetoresistance of C_I . The R_{Iz} and R_{IS} can be calculated as:

$$\begin{cases} R_{Iz} = \frac{t}{\mu_{vz}(a-2t)} \\ R_{IS} = \frac{b-2t}{\mu_0(a-2t)} \end{cases} \quad (23)$$

Based on Eq. (18) to Eq. (23), the following can be derived:

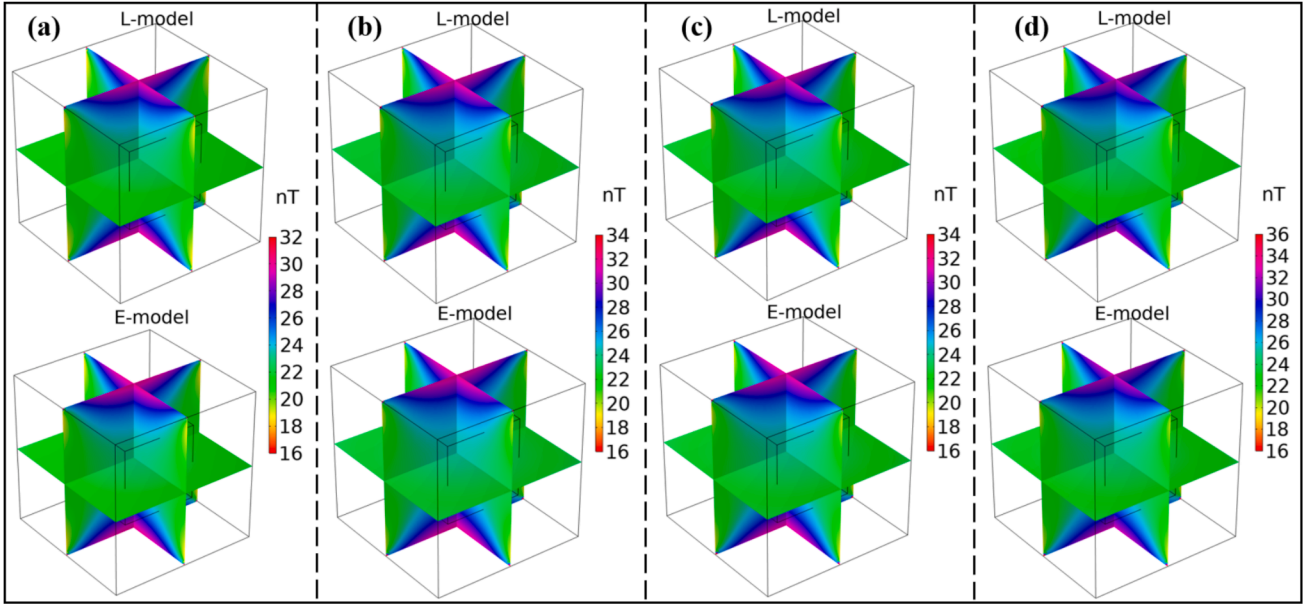


Fig. 15. The distribution of remanence for L-model compared to E-model with different frequencies. (a) 25 Hz, (b) 50 Hz, (c) 75 Hz, (d) 100 Hz.

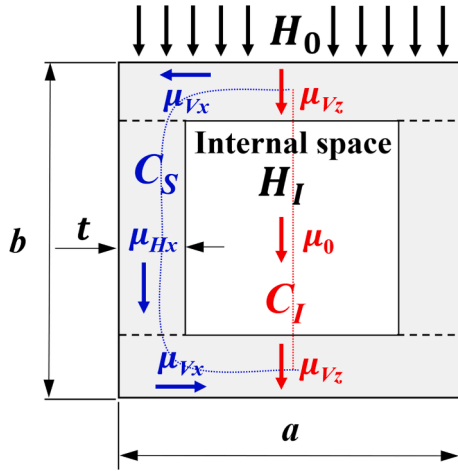


Fig. 16. The equivalent magnetic circuit of PN-LCSD.

$$\begin{cases} U_s = \left(\frac{\mu_{Hx}a}{\mu_{Vx}} + b - 2t \right) H_s \\ U_t = \left(\frac{2\mu_0 t}{\mu_{Vz}} + b - 2t \right) H_t \end{cases} \quad (24)$$

Since $U_s = U_t$, and further applying Eq. (19) and Eq. (24), the SF of PN-LCSD can be expressed as:

$$SF = \frac{H_0}{H_I} = \frac{4\mu_0\mu_{Vx}\mu_{Hx}t^2 + 2\mu_{Vx}\mu_{Vz}\mu_{Hx}(b-2t)t + \frac{a-2t}{a}}{\mu_0\mu_{Vz}\mu_{Hx}a^2 + \mu_0\mu_{Vx}\mu_{Vz}a(b-2t)} \quad (25)$$

Due to $a, b \gg t$, Eq. (25) can be approximated as follows:

$$SF = \frac{H_0}{H_I} = \frac{4\mu_0\mu_{Vx}\mu_{Hx}t^2 + 2\mu_{Vx}\mu_{Vz}\mu_{Hx}bt}{\mu_0\mu_{Vz}\mu_{Hx}a^2 + \mu_0\mu_{Vx}\mu_{Vz}ab} + 1 \quad (26)$$

Using Eq. (25), the SF of PN-LCSD is calculated for an internal space dimension of 400 mm × 400 mm × 400 mm at values of $p = 0.5$ ($d_B = 1$ mm, $d_T = 2$ mm), $p = 1$ ($d_B = d_T = 2$ mm), $p = 1.5$ ($d_B = 1.5$ mm, $d_T = 1$ mm) and $p = 2$ ($d_B = 2$ mm, $d_T = 1$ mm). The comparison of calculated value and simulative value for SF is shown in Table 4, and the maximum deviation is only 0.84 %. The comparison results verify the effectiveness

Table 4

The comparison of simulated and calculated SF for PN-LCSD at different thickness ratios.

p	SF (simulation)	SF (calculation)	Deviation (%)
0.5	392.62	390.94	0.43
1	551.88	547.27	0.84
1.5	364.30	361.44	0.79
2	448.83	445.94	0.64

of the SF calculation equation.

5. Experimental verification

The shielding performance test system for PN-LCSD is shown in Fig. 17. The inner support structure of the PN-LCSD used in the experiment is constructed by welding 2 mm thick 1J85 permalloy plates, forming an internal space of 400 mm × 400 mm × 400 mm. The preparation process of 1J85 permalloy plates primarily includes raw material preparation, melting, casting, hot rolling, cold rolling, heat treatment, surface treatment, and inspection. Subsequently, the 1K107B nanocrystalline film (20 μm / layer, a total of 40 layers) is successively laminated onto the permalloy surface using AS-DC10X-BC double-sided adhesive (10 μm thick) produced by ASTECH. The adhesive consists of a central PET transparent film and acrylic adhesive layers on both sides, providing strong adhesion and non-magnetism. The 1K107B nanocrystalline film used in the experiment is an iron-based nanocrystalline alloy, primarily composed of Fe, Si, B, Nb, and Cu. The preparation process of the 1K107B nanocrystalline film includes ribbon spraying, winding, annealing, coating, and lamination. A through-hole with a diameter of 20 mm is centrally located on the cover plate of the PN-LCSD for circuit layout.

A pair of square coils along the x direction in the three-axis coil system are selected as the test coils, with 30 turns and dimensions of 1.68 m × 1.68 m, and a coil spacing of 0.91 m. Before starting the test, a handheld demagnetizer is used to demagnetize the PN-LCSD and the flux gate, signal generator, power amplifier, and current detector are turned on for preheating. The current generated by the signal generator is amplified by the power amplifier and then passed to the test coil, while the current detector is used to monitor the current in real-time. The magnetic field amplitude produced by the coil is changed through

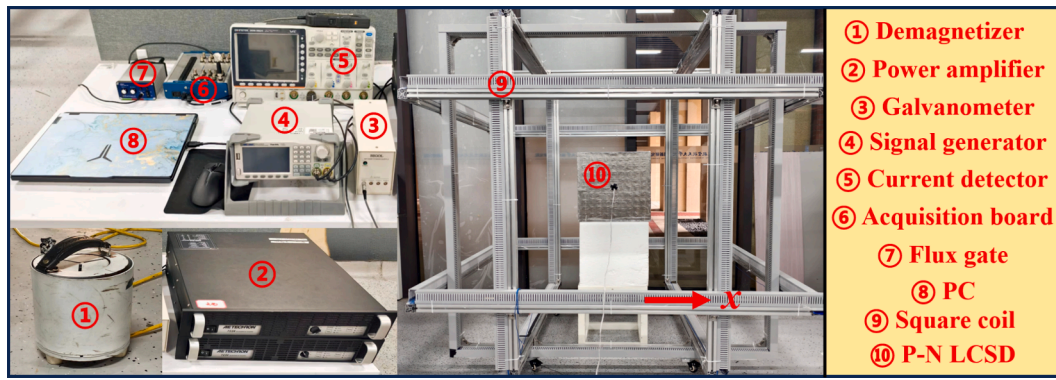


Fig. 17. Shielding performance test system of PN-LCSD.

adjusting the current during testing.

The shielding performance of PN-LCSD in alternating magnetic fields is influenced by both the magnetic properties and electrical conductivity of the permalloy and nanocrystalline. Since this paper focuses on the equivalent magnetic properties of PN-LCSD, the static magnetic fields are selected as the experimental background fields to avoid interference from eddy current effects caused by the material conductivity. As shown in Fig. 18, the device is tested in 10 groups (Group1 to Group10) in the range of approximately 5000 to 50,000 nT. In section 2.1, the magnetic properties of 1J85 permalloy and 1K107B nanocrystalline are measured at a controlled room temperature of 25°C. To prevent temperature from affecting the magnetic properties of the materials and thus interfering with the measured results, all 10 groups of tests are also carried out at 25°C. In each group of tests, the flux gate is placed at the center of the coil to detect the environmental magnetic field along the x direction, then the device is placed at the center of the coil to detect the remanence at the center of the internal space.

According to the initial permeability of 1J85 and 1K107B and the μ of PN-LCSD used in the experiments, using Eq. (17), the equivalent permeability of the device can be obtained as: $\mu_{Vx} = \mu_{Vy} = 49.02$ k, $\mu_{Hx} = \mu_{Hy} = 56.50$ k, $\mu_{Vz} = \mu_{Hz} = 54.69$ k. Using the equivalent permeability of PN-LCSD used in the experiments as the input parameter and the static magnetic fields from the 10 groups of tests as the background fields, the simulation model is constructed in COMSOL software based on the equivalent surface method to simulate its internal remanence. The simulated remanence distribution of PN-LCSD used in the experiments from Group (1) to Group (10) is shown in Fig. 19. The comparison

between the simulated and measured remanence is shown in Fig. 20(a). Compared to the measured remanence, the deviations of the simulation values across the 10 tests (Group1 to Group10) range from 3.68 % to 6.48 %. The comparison results verify the accuracy of the equivalent permeability calculation model for PN-LCSD.

Based on the equivalent permeability, combined with Eq. (25), the SF of PN-LCSD used in the experiments is calculated to be 360.70. The comparison between the measured and calculated values of SF for the device from Group (1) to Group (10) is shown in Fig. 20(b). Compared to the measured SF, the deviations of the calculated values across the 10 tests (Group1 to Group10) range from 2.69 % to 5.78 %. The main reason for the measured performance being slightly lower than the theoretical performance is that the center hole and small gaps cause partial magnetic leakage. The experimental results further demonstrate the accuracy and effectiveness of the calculation method for the equivalent magnetic properties of PN-LCSD.

6. Conclusion

This paper proposes a novel theoretical model for the equivalent magnetic properties of PN-LCSD which can calculate the equivalent permeability and SF of PN-LCSD, and its effectiveness is verified by the simulations and experiments. Using the equivalent permeability of PN-LCSD, combined with the equivalent surface method, the problems of mesh generation and convergence caused by the multi-thin layer characteristic and magnetic circuit complexity in its performance analysis can be solved. Based on the equivalent permeability, the proposed model further provides the analytical SF of PN-LCSD, which is used to evaluate its shielding performance. The proposed model provides a valuable theoretical basis for the performance analysis and design optimization of PN-LCSD.

However, considering the potential challenges in the industrial application of PN-LCSD, there is still room for further improvement of the proposed model. When the volume and shielding layer thickness of PN-LCSD expand, the attachment area and layer number of nanocrystalline films and the welding difficulty of permalloy will increase, which further aggravates the magnetic leakage gaps and installation stress of PN-LCSD, thus reducing its shielding performance. Therefore, the PN-LCSD is not suitable for large-scale construction and is mainly used in medium or small shielding systems. In conclusion, establishing a theoretical model for the equivalent magnetic properties of PN-LCSD considering the influence of magnetic leakage gaps and installation stress can further improve the accuracy and effectiveness of the proposed model to meet a wider range of analysis and application requirements.

7. Discussion

The Eq. (17) and Eq. (25) demonstrate that the proposed theoretical

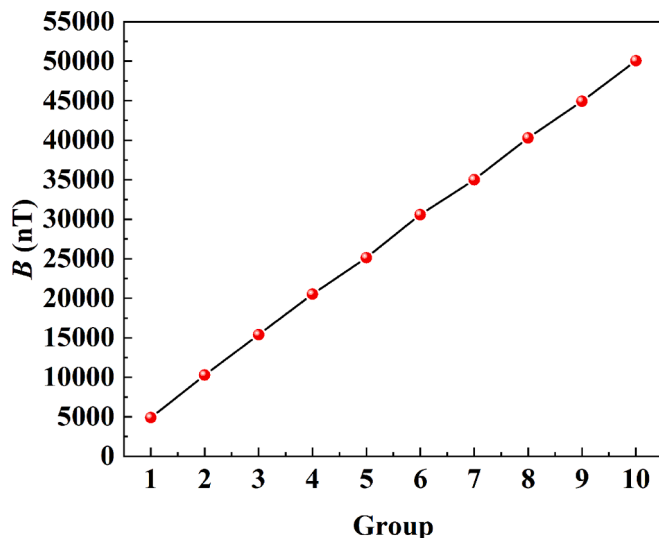


Fig. 18. The environmental field strength of Group (1) to Group (10).

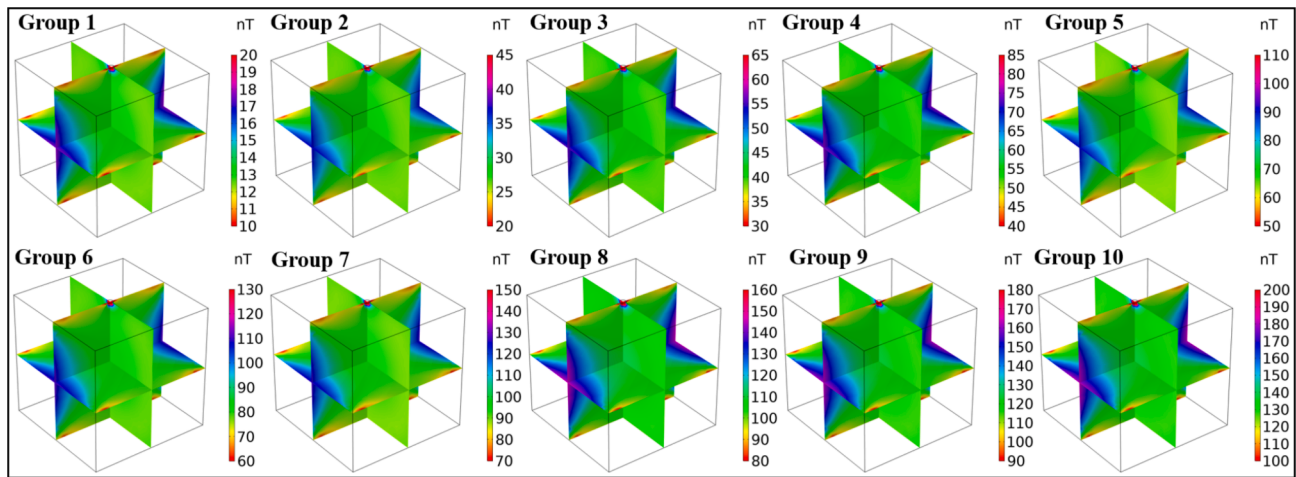


Fig. 19. The simulated remanence distribution of PN-LCSD from Group (1) to Group (10).

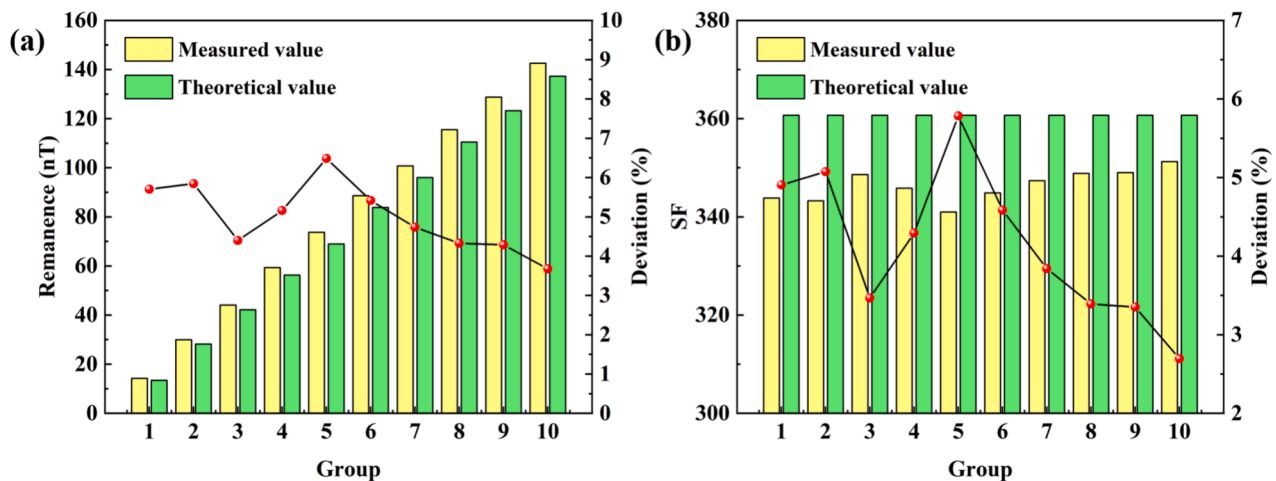


Fig. 20. The comparison between the measured and theoretical values of the shielding performance for PN-LCSD. (a) Remanence, (b) SF.

model for equivalent magnetic properties is constructed based on material properties and geometric parameters, which is not limited to PN-LCSD analysis but applicable to all laminated composite shielding device (LCSD). By inputting the material permeabilities and dimensions of composite shielding layers, the model can calculate the equivalent permeability and SF of LCSD, enabling the evaluation of its shielding performance.

The construction of LCSD require a comprehensive consideration of cost and performance metrics based on the application requirement. Besides the permalloy and nanocrystalline, high-permeability silicon steel and low-loss ferrite are used in some specific shielding devices. High-permeability silicon steel exhibits lower cost, permeability, mechanical properties, and weldability compared to the permalloy [23]. Substituting the permalloy with high-permeability silicon steel can reduce the cost of LCSD, but it results in a degradation of its shielding performance. Furthermore, low-loss ferrite demonstrates lower conductivity, permeability, and mechanical properties than the nanocrystalline [24,25]. Replacing the nanocrystalline with low-loss ferrite can reduce magnetic noise in LCSD to some extent, but this substitution leads to a deterioration in its shielding performance, accompanied by an increase in its weight and structural complexity. In conclusion, when the practical demand for shielding performance is low, the high-permeability silicon steel and low-loss ferrite may also be considered as alternative materials for PN-LCSD to meet specific requirements for cost reduction or magnetic noise minimization.

CRediT authorship contribution statement

Xueping Xu: Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Lei Wang:** Writing – original draft, Visualization, Validation, Investigation. **Wei Liu:** Visualization, Software, Investigation, Formal analysis. **Zhenkai Zhao:** Visualization, Validation, Software, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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